

Feature A (Stress):

Chosen values / Knowns:

- $W = \text{distributed load} = 2F$
- FOS (Factor of safety) = 4
- (Chosen) • $F = 500 \text{ lbf}$
- (Chosen) • Material: Aluminum 6061-T6
- $\sigma_{\text{yield}} = 40000 \text{ psi}$ (Matweb)
- $L_A = 1 \text{ in}$ (From 2157 TOPICS

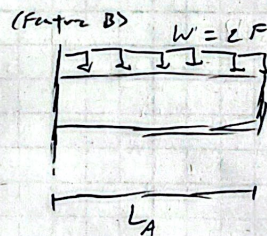
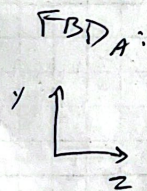
"The hole in the link that connects to the 1-inch diameter shaft...")

Unknowns:

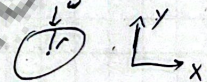
- Section Modulus (Z) (in^3)
- Minimum radius (r) (in)
- Moment (M_A)

Assumptions:

- Feature A acts as cantilever beam fixed at connector bar (Feature B)
- Cantilever beam experiences unknown distributed load (W)



Cross-section A-A:



Solving For Z :
(Section modulus)

$$\sigma_{\text{allow}} = \frac{W L_A}{2 \cdot Z} \quad (\text{eq. 1})$$

$$\sigma_{\text{allow}} = \frac{\sigma_{\text{yield}}}{FOS}$$

$$\frac{W L_A}{2 Z} = \frac{\sigma_{\text{yield}}}{FOS}$$

(From σ_{allow}): $Z = \frac{FOS (2F) (L_A)}{2 \sigma_{\text{yield}}}$

Solving For r :

(From Geometry): $r = \sqrt[3]{\frac{4(Z)}{\pi}} \rightarrow Z = \frac{\pi r^3}{4}$
 = (eq. 2)

Feature A (Stress) Cont.

$$(\sigma_{max}): Z = \frac{F_o(2F)(L_A)}{2(\sigma_{yield})}$$

$$(\text{geometry}): Z = \frac{\pi \cdot r^3}{4}$$

$$\frac{F_o(2F)(L_A)}{2(\sigma_{yield})} = \frac{\pi \cdot r^3}{4}$$

$$r_{A(stress)} = \sqrt[3]{\frac{2F_o(2F)(L_A)}{\pi(\sigma_{yield})}}$$

$$r_{A(stress)} = \sqrt[3]{\frac{2(4)(2.500\text{ lbf})(1\text{ in})}{\pi(40000\text{ psi})}}$$

$$\left| \begin{array}{l} r_{A(stress)} = 0.399\text{ in} \\ d_{A(stress)} = 0.798\text{ in} \end{array} \right| \rightarrow d_{A(stress)} = 2 \cdot r_A$$

Feature A (Stiffness):

Knowns:

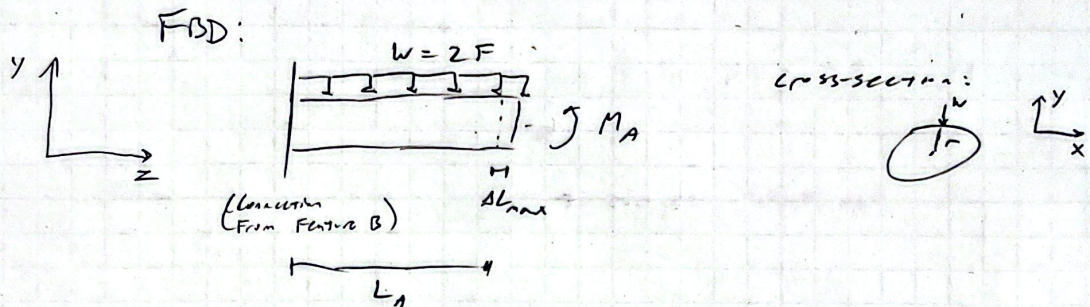
- F (Applied Force) = 500 lbf
- W = distributed load = $2F = 1000\text{ lbf}$
- $\Delta L_{max} = 0.005\text{ in}$
- Elasticity (E) = 10000 ksi
= $10 \cdot 10^6\text{ psi}$ (MATLAB)
- $L_A = 1\text{ in}$

Unknowns:

- I_A (moment of inertia)
- $r_{stiff_{min}}$ or $d_{stiff_{min}}$

No new Assumptions Beside ones in Feature A (Stress)

Feature A (Stiffness) cont.



From Beam Tables, Fixed cantilever with uniform W :

$$\Delta L = \frac{WL^4}{8EI} \quad (\text{eq. 1})$$

$$\Delta L_{\max} = \frac{2F(L_A)^4}{8EI_A}$$

$$(\text{From } \Delta L_{\max}): I_A = \frac{2F(L_A)^4}{8E\Delta L_{\max}}$$

$$(\text{From geometry}): I_A = \frac{\pi}{64} d_A^4$$

$$\frac{2F(L_A)^4}{8E\Delta L_{\max}} = \frac{\pi}{64} d_A^4$$

$$d_{A \text{ stiff min}} = \sqrt[4]{\frac{2F(L_A)^4 \cdot 64}{8 \cdot \pi E (\Delta L_{\max})}}$$

$$d_{A \text{ stiff min}} = \sqrt[4]{\frac{2(500 \text{ N})(1 \text{ m})^4 \cdot 64}{8 \cdot \pi (10 \cdot 10^6 \text{ psi})(0.005 \text{ m})}}$$

$$d_{A \text{ stiff min}} = 0.475 \text{ in}$$

Governing d_A :

$$d_{A \text{ stress min}} > d_{A \text{ stiff min}}$$

$$d_A = 0.798 \text{ in}$$

Feature B (Stress):

Knowns:

- Axial Pull Force (P) = $2F = 2(500 \text{ lb}_f) = 1000 \text{ lb}_f$
- $FOS = 4$
- $\sigma_{\text{yield}} = 40000 \text{ psi}$ (MatWeb)

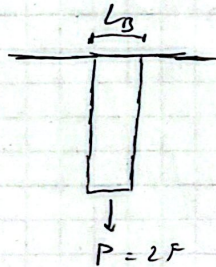
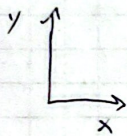
Unknown:

- σ_{allow}
- A_B (cross-sectional Area)
- T_B (Thickness)

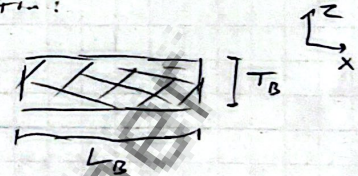
Assumptions:

- Feature B acts as Axial loaded bar
- Force P pulls uniformly in cross-section
- Assume L_B (length) is same as nominal dimension "a" in Figure 1
($a = 0.498 \text{ in}$)

FBD:



Cross-section:



Solving For A_B :

$$A_B = L_B \cdot T_B$$

Solving For σ_{allow} :

$$\sigma_{\text{allow}} = \frac{\sigma_{\text{yield}}}{FOS}$$

$$\sigma_{\text{allow}} = \frac{F}{A} = \frac{P}{A_B} = \frac{P}{L_B \cdot T_B} = \frac{2F}{L_B \cdot T_B}$$

$$\frac{\sigma_{\text{yield}}}{FOS} = \frac{2F}{L_B \cdot T_B}$$

Feature B (Stress) calc:

Solving For T_B :

$$\frac{\sigma_{yield}}{FOS} = \frac{2F}{L_B \cdot T_B}$$

$$T_{B \text{ stress}} = \frac{2F(FOS)}{\sigma_{yield} \cdot L_B}$$

$$T_{B \text{ stress}} = \frac{2(500 \text{ lbf})(4)}{(40000 \text{ psi})(0.498 \text{ in})}$$

$$T_{B \text{ stress}} = 0.201 \text{ in}$$

Feature B (Stiff):

Known:

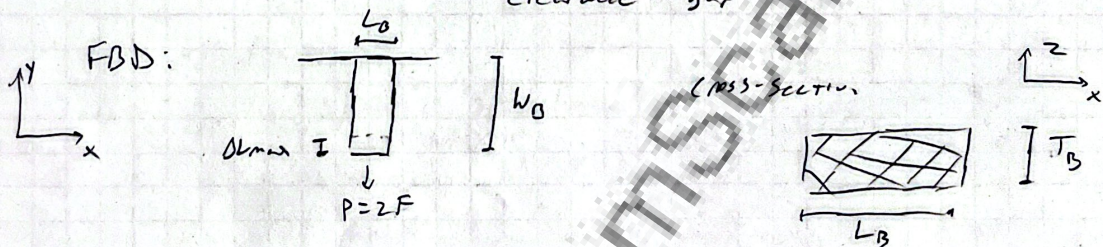
- Pull Force (P) = $2F = 1000 \text{ lbf}$
- $\Delta L_{max} = 0.005 \text{ in}$
- Elasticity (E) = $10 \cdot 10^6 \text{ psi}$
- $L_B = 0.498 \text{ in}$ (From Assumption)

Unknowns:

- $T_B \text{ stiff}$
- A_B (cross-sectional Area)

New Assumptions:

- Assumed width (W_0) = 0.75 in Based off half of total structural horseshoe profile "C" (1.499 in) to bridge clearance gap



Solving For A_B :

$$A_B = L_B \cdot T_B$$

Feature B (Stiff) case:

Bending Formula: $\Delta L = \frac{PL}{AE}$

$$\Delta L_{max} = \frac{2F(W_B)}{(T_B \cdot L_B)}$$

$$T_{B \text{ STIFF}} = \frac{2F(W_B)}{L_B \cdot \Delta L_{max} \cdot E}$$

$$T_{B \text{ STIFF}} = \frac{2(50016\text{N})(0.75\text{m})}{(0.498\text{m})(0.005\text{m})(10 \cdot 10^6\text{psi})}$$

$$T_{B \text{ STIFF}} = 0.0301\text{m}$$

Determining Governing Thickness (T_B)

$$T_{B \text{ STRESS}} > T_{B \text{ STIFF}}$$

$$T_B = 0.201\text{m}$$

Feature C (Stress)

Knowns:

• Concentrated Center load ($P = 2F$)

• $F_{US} = 4$

• $\sigma_{yield} = 40000\text{psi}$ (Material)

• $T_C = 1\text{in}$ (From given Assumption)

Unknowns:

• Max Bending moment (M_{max})

• Section Modulus (Z)

• Width (W_C)

Assumptions:

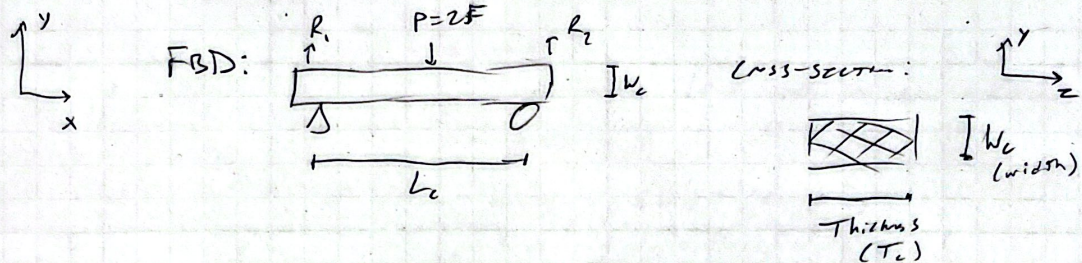
• Feature C behaves as simply supported beam
; concentrated load ($P = 2F$) AT mid span
From Feature B

• Length (L_C) can be determined From nominal dimensions

$$L_C = a + 2b + \text{offset parameter}$$

$$L_C = (0.498\text{m}) + 2(0.9992\text{m}) + 0.10\text{m} = 2.5964\text{m}$$

Feature C (Stress) cas:



Due to symmetry:

$$R_1 = R_2 = F = 5001b_k$$

Maximum Internal Bending moment in simply supported beam with concentrated load at midspan:

$$M_{max} = \frac{PL}{4} \quad (\text{eq 1})$$

$$M_{max} = \frac{2F(L_c)}{4}$$

Section Modulus (Z) for rectangular cross-section:

$$Z = \frac{bh^2}{6} \quad (\text{eq 2})$$

$$Z = \frac{T_c(W_c)^2}{6}$$

Solving For Tension:

$$\sigma_{allow} = \frac{Mc}{I} = \frac{M_{max}}{Z} \quad (\text{eq 1})$$

$$\sigma_{allow} = \frac{\sigma_{yield}}{FOS}$$

$$\frac{M_{max}}{Z} = \frac{\sigma_{yield}}{FOS}$$

$$\frac{\left[\frac{2F(L_c)}{4} \right]}{\left[\frac{T_c(W_c)^2}{6} \right]} = \frac{\sigma_{yield}}{FOS}$$

$$W_c \text{ stress} = \sqrt{\frac{1.5(2F)L_c \cdot FOS}{T_c \cdot \sigma_{yield}}}$$

$$W_c \text{ stress} = \sqrt{\frac{1.5(2 \cdot 5001b_k)(2.5464m)(4)}{(1m)(40000psi)}}$$

$$W_c \text{ stress} = 0.624m$$

Feature C (Stiffness)

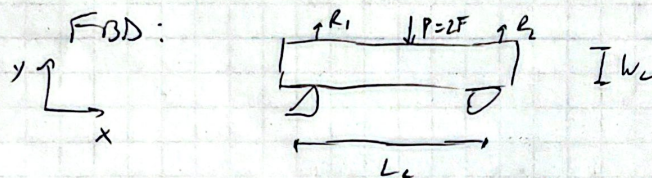
Knowns:

- Concentrated load at midspan ($P = 2F$)
- $F_{03} = 4$
- Elasticity (E) = $10 \cdot 10^6$ psi
- $L_c = 2.5964$ m
- $T_c = 1$ m (From overall Thickness Assumption)

Unknown:

- Moment of Inertia (I_c)
- Width (W_c)

New Assumptions: None

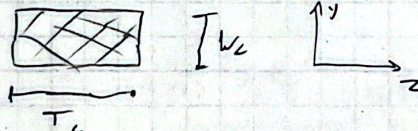


Due to Symmetry:

$$R_1 = R_2 = F = 500 \text{ lbf}$$

$$P = 2(500 \text{ lbf}) = 1000 \text{ lbf}$$

Cross-section:



Standard Equation For Simply Supported beam
with P Force at midspan:

$$\Delta L = \frac{P L^3}{48 E I} \quad (\text{eq. 1})$$

$$\Delta L_{\max} = \frac{2F(L_c)^3}{48 E I_c}$$

Solving For I_c :

$$I_c = \frac{2F(L_c)^3}{48 E \cdot \Delta L_{\max}} \quad (\text{eq. 2})$$

I (moment of Inertia) For cross-section rectangle: $I = \frac{1}{12} b h^3$

$$I_c = \frac{T_c (W_c)^3}{12}$$

Feature (Stiffness) eq:

$$I_{\text{(From NIS section)}} = \frac{T_c (W_c)^3}{12}$$

$$I_{\text{(From Beam eqn)}} = \frac{2F(L_c)^3}{48 \cdot E \cdot \Delta L_m}$$

$$\frac{T_c (W_c)^3}{12} = \frac{2F(L_c)^3}{48 \cdot E \cdot \Delta L_m}$$

Solving For W_c :

$$W_{c \text{ start}} = \sqrt[3]{\frac{12(2F)(L_c)^3}{48(E)(\Delta L_m)(T_c)}}$$

$$W_{c \text{ start}} = \sqrt[3]{\frac{12(2.50016 \text{ N})(2.5967 \text{ m})^3}{48(10 \cdot 10^6 \text{ psi})(0.005 \text{ m})(1 \text{ N})}}$$

$$W_{c \text{ start}} = 0.449 \text{ m}$$

Determining Governing W_c :

$$W_{c \text{ (stress)}} > W_{c \text{ (stiff)}}}$$

$$W_c = 0.624 \text{ m}$$

Feature D (Stress):

Knowns:

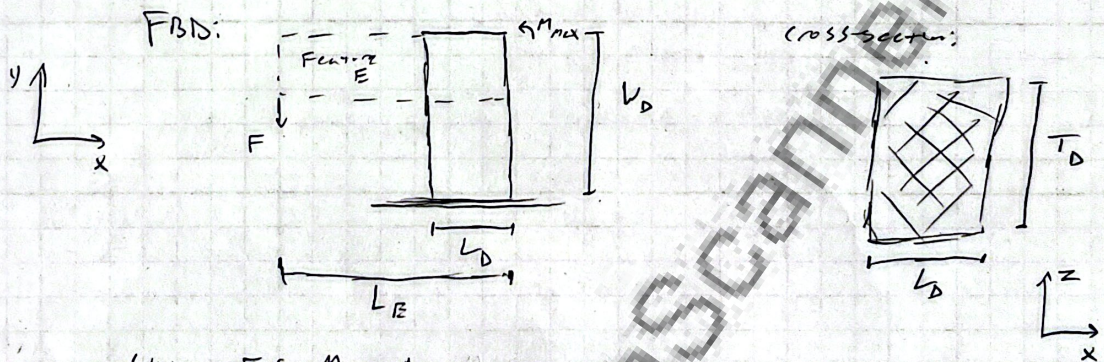
- Reaction load Per Side due to symmetry of overall shape:
 $F = 50016x$
- $FOS = 4$
- $\sigma_{yield} = 40000 \text{ psi}$
- $T_D = 1in$ due to Thickness Assumption

Unknowns:

- Bending moment From Feature E (M_{max})
- Length (L_D)
- Z_D (Section modulus)
- Tension

Assumptions:

- Feature D acts as vertical cantilever beam Fixed rigidly at connection to Feature C
- M_{max} is cause by Feature E $\rightarrow$ use moment arm (length of E)
- Assume width From nominal dimension ($"b" = 0.9992in$)
(W_D)
- Assume $L_E = (\text{Nominal } "b") = 0.9992in$



Solving For M_{max} :

$$M_{max} = F \cdot L_E$$

Feature D (Stress) calc:

Solving For Section Modulus (Z):

(Formula for rectangle cross-section): $Z = \frac{bh^2}{6}$ (eq 1)

$$Z = \frac{T_D (L_D)^2}{6}$$

Solving For Stress:

$$\sigma_{allow} = \frac{M_L}{I} = \frac{M_{max}}{Z} \quad (\text{eq 2})$$

$$\sigma_{allow} = \frac{(F \cdot L_E)}{\left[\frac{T_D (L_D)^2}{6} \right]}$$

Solving For L_D :

$$L_{D_{stress}} = \sqrt{\frac{6 \cdot F \cdot L_E \cdot F_{os}}{T_D (\sigma_{allow})}}$$

$$L_{D_{stress}} = \sqrt{\frac{6 (500 \text{ lb}_f) (0.9992 \text{ in}) (4)}{(1 \text{ in}) (40000 \text{ psi})}}$$

$$L_{D_{stress}} = 0.5475 \text{ in}$$

Feature D (Stiffness):

Knowns:

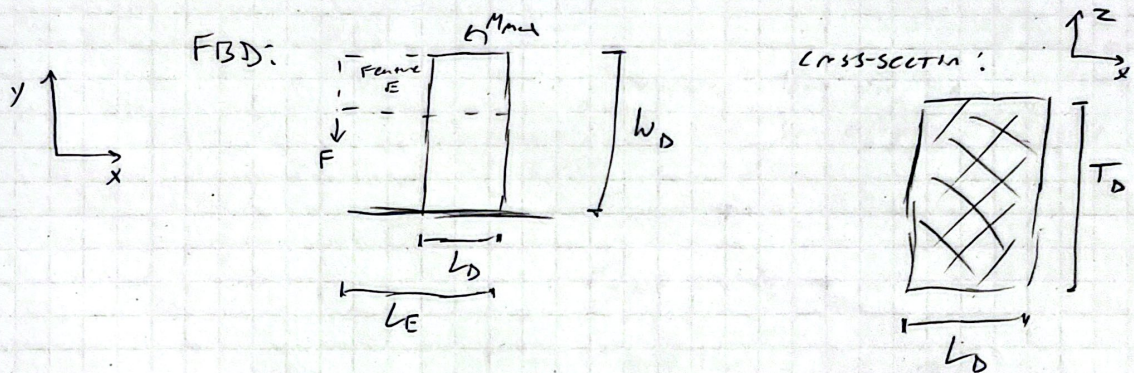
- Load per side due to symmetry: $F = 500 \text{ lb}_f$
- Width (W_D) = 0.9992 in
- $\Delta L_{max} = 0.005 \text{ in}$
- Elasticity (E) = $10 \cdot 10^6 \text{ psi}$
- $T_D = 1 \text{ in}$

Unknowns:

- Moment of Inertia (I)
- Length (L_D)

New Assumptions: None

Fenetic (D) (Stiffness)



Solving For $M_{max} = F \cdot L_E$

Vertical cantilever beam with moment at edge:

$$\Delta L = \frac{ML^2}{2EI} \quad (\text{eq 1})$$

$$\Delta L_{max} = \frac{M_{max} W_D^2}{2EI} = \frac{(F \cdot L_E)(W_D)^2}{2EI_D}$$

Moment of Inertia (I) From cross-section:

(eq 2)
$$I_D = \frac{T_D (L_D)^3}{12} \quad \left(\begin{array}{l} \text{due to anisotropy} \\ \text{of moment} \\ \therefore \text{vertical} \\ \text{cantilever} \end{array} \right)$$

Solving For L_D :

$$\Delta L_{max} = \frac{F(L_E)(W_D)^2}{2E \left(\frac{T_D (L_D)^3}{12} \right)}$$

$$L_{D \text{ STIFF}} = \sqrt[3]{\frac{6(F)(L_E)(W_D)^2}{E(T_D)(\Delta L_{max})}}$$

$$L_{D \text{ STIFF}} = \sqrt[3]{\frac{6(52016)(0.9992 \text{ in})(0.9992 \text{ in})^2}{(10.106 \text{ psi})(1 \text{ in})(0.005 \text{ in})}}$$

$$L_{D \text{ STIFF}} = 0.391 \text{ in}$$

Determining Geometry L_D :

$$L_{D \text{ STRESS}} > L_{D \text{ STIFF}}$$

$$L_D = 0.5475 \text{ in}$$

Feature E (stress):

Known:

- Transferred load per side due to symmetry: $F = 500 \text{ lb}$
- $FOS = 4$
- $\sigma_{\text{yield}} = 40000 \text{ psi}$
- $T_E = 1 \text{ in}$ (depth assumption)
Symmetry:

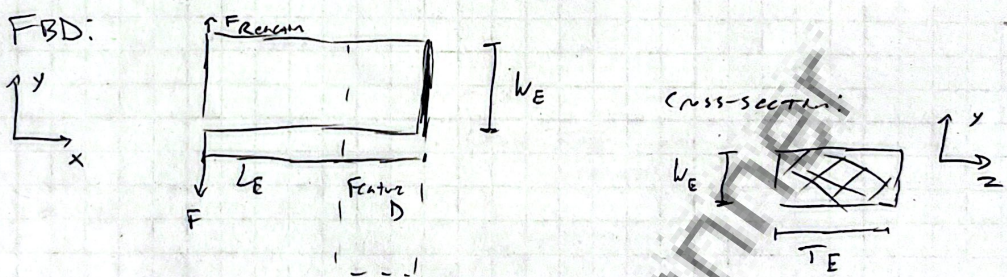
Unknown:

- Max Bending Moment (M_{max})
- Section modulus (Z)
- Minimum width (W_E)

Assumptions:

- Feature E acts as a horizontal cantilever beam, fixed at connection to Feature D
- Assume length based on nominal dimension
"b" = 0.9992 in

FBD:



M_E based on F :

$$M_E = F(L_E)$$

Solving for Section modulus:

(Z) Section modulus Based on cross-section:

$$Z = \frac{b h^2}{6} = \frac{T_E (W_E)^2}{6}$$

Feature (E) (stress) calc:

Solving For σ_{allow} :

$$\sigma_{allow} = \frac{\sigma_{yield}}{FOS} \quad (\text{eq 1})$$

$$\sigma_{allow} = \frac{M_C}{I} = \frac{M_E}{Z} \quad (\text{eq 2})$$

Solving For W_E :

$$\frac{\sigma_{yield}}{FOS} = \frac{M_E}{Z}$$

$$\frac{\sigma_{yield}}{FOS} = \frac{[F(L_E)]}{\left[\frac{T_E (W_E)^2}{6} \right]}$$

$$W_{E \text{ stress}} = \sqrt{\frac{6(FOS \cdot F(L_E))}{T_E \cdot \sigma_{yield}}}$$

$$W_{E \text{ stress}} = \sqrt{\frac{6(4)(500 \text{ lb}_f)(0.9992 \text{ m})}{(1 \text{ m})(40000 \text{ psi})}}$$

$$W_{E \text{ stress}} = 0.5475 \text{ m}$$

Feature E (stiff) ?

Knowns:

- Load Force per side based on symmetry: $F = 500 \text{ lb}_f$
- $\Delta L_{max} = 0.005 \text{ m}$
- Elasticity (E) = $10.12 \times 10^6 \text{ psi}$
- $T_E = 1 \text{ m}$

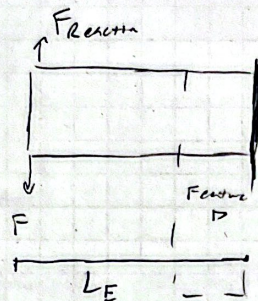
Unknowns:

- Moment of Inertia (I_E)
- Width (W_E)

New Assumptions: None

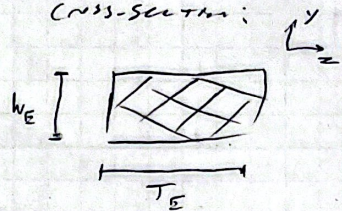
Feature E (START) cond:

FBD:



L_E

Cross-section:



Cantilever Beam with concentrated load at Free End:

$$\Delta L = \frac{PL^3}{3EI} \quad (\text{eq 1})$$

$$\Delta L_{max} = \frac{F(L_E)^3}{3EI}$$

Solving For Moment of Inertia (I_E):

$$I_E = \frac{F(L_E)^3}{3E \cdot \Delta L_{max}}$$

Solving For (I_E) Through cross-section:

$$I_E = \frac{1}{12} b h^3 \quad (\text{eq 2})$$

$$I_E = \frac{T_E (W_E)^3}{12}$$

Solving For W_E :

$$\frac{F(L_E)^3}{3E \cdot \Delta L_{max}} = \frac{T_E (W_E)^3}{12}$$

$$W_{E \text{ START}} = \sqrt[3]{\frac{12(F)(L_E)^3}{3E \cdot \Delta L_{max} \cdot T_E}}$$

$$W_{E \text{ START}} = \sqrt[3]{\frac{12(500 \text{ lb}_f)(0.9992 \text{ m})^3}{3(10 \cdot 10^9 \text{ psi})(0.005 \text{ in})(1 \text{ in})}}$$

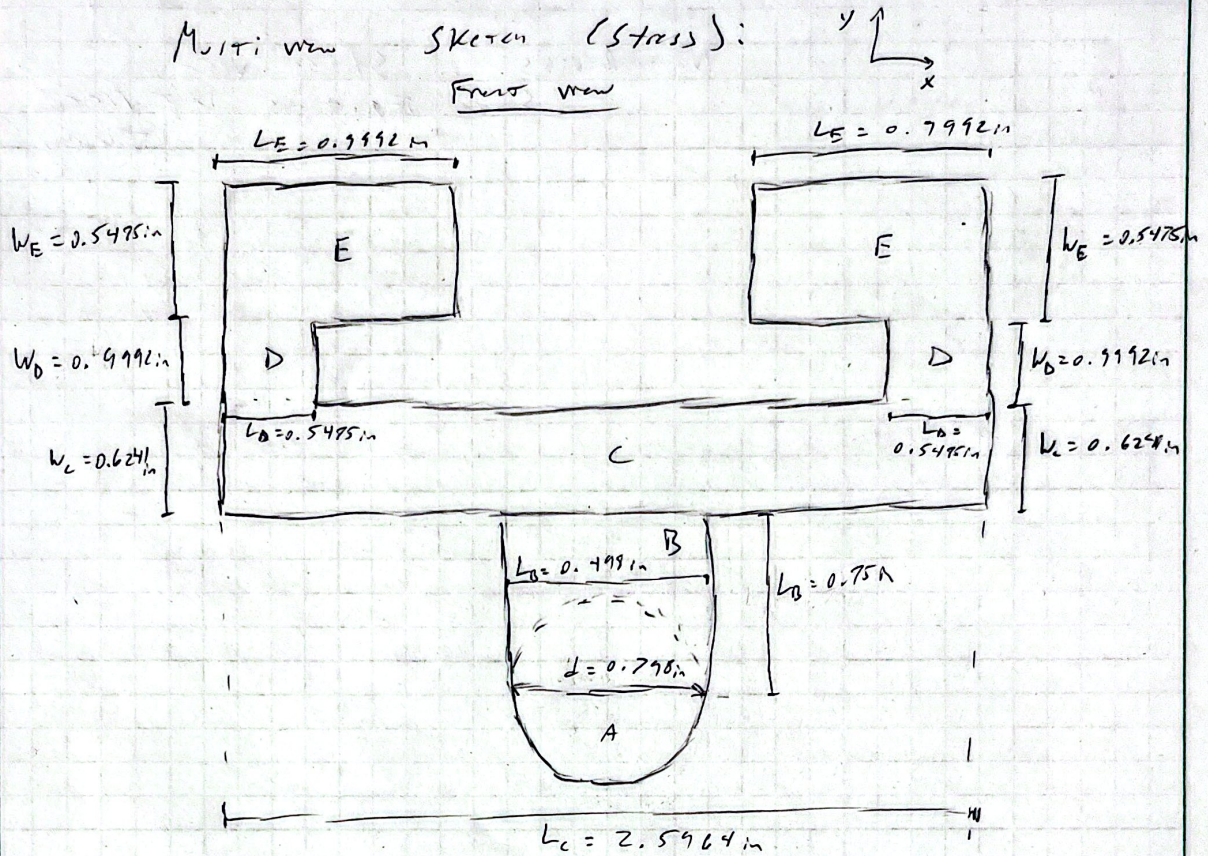
$$W_{E \text{ START}} = 0.392 \text{ m}$$

Determining Governing W_E :

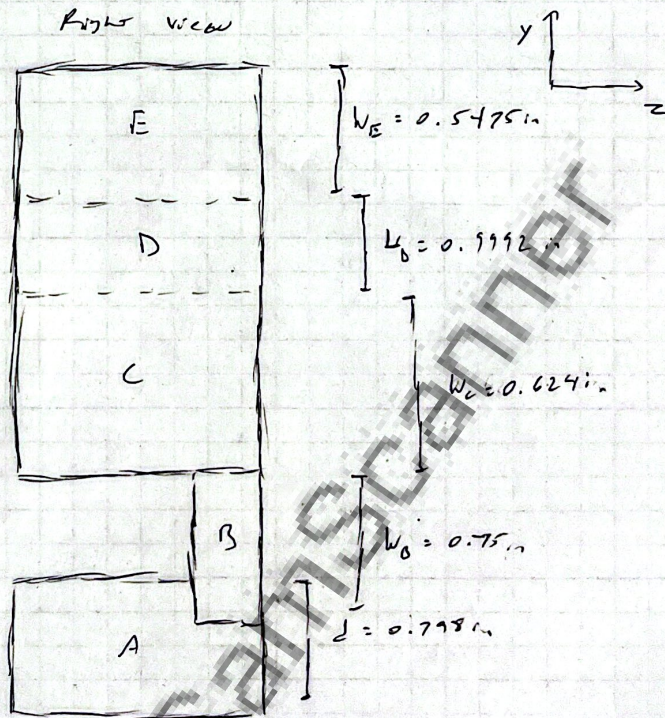
$$W_{E \text{ stress}} > W_{E \text{ START}} \rightarrow W_E = 0.548 \text{ m}$$

Multi view Sketch (stress):

Front view



Right view

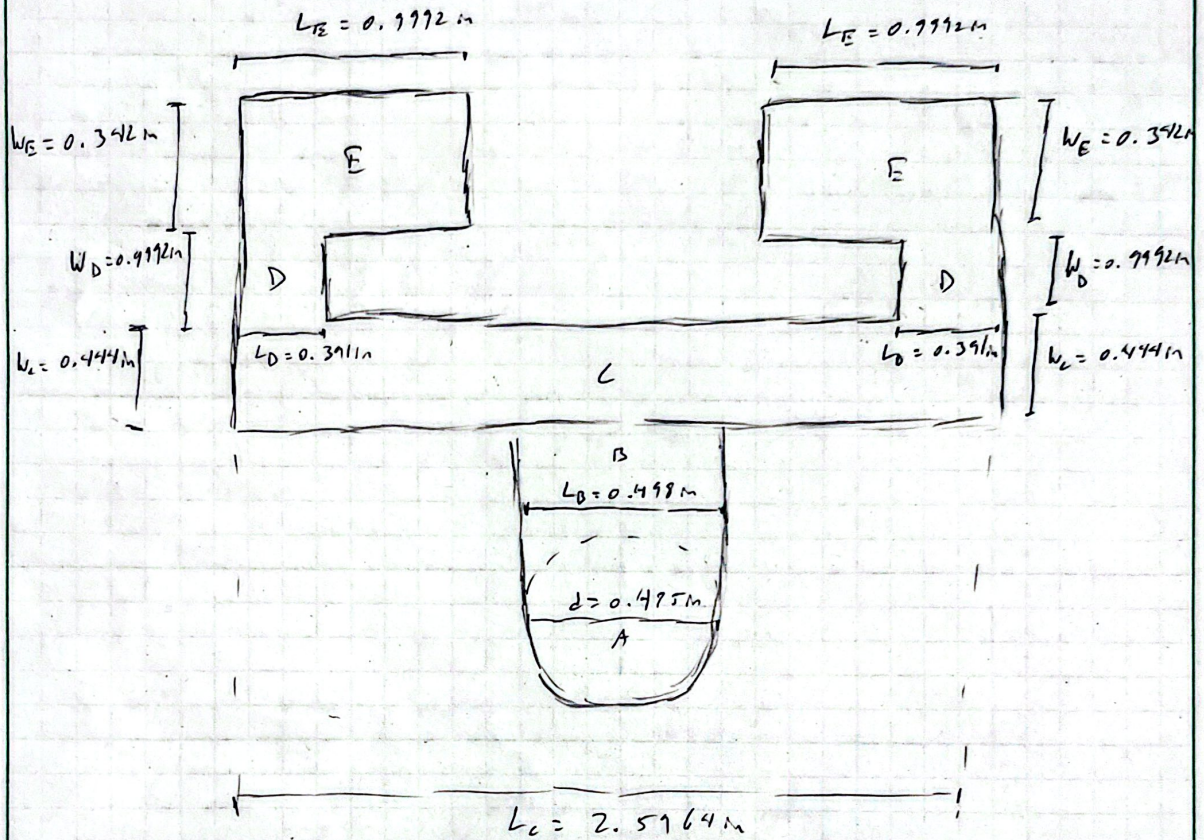


Thickness is (due to Assumption)

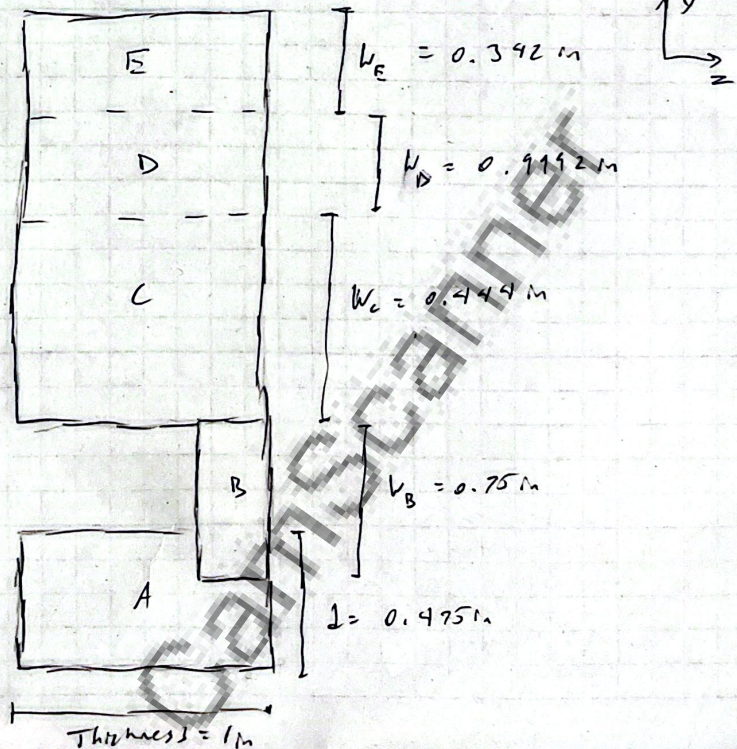
Multi view Sketch (Sketch):

$\begin{matrix} y \\ \nearrow \\ L_x \end{matrix}$

Front View

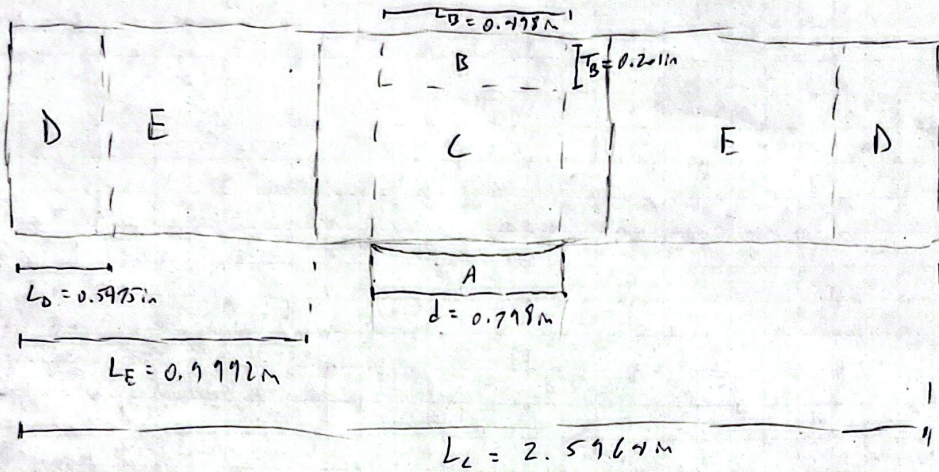
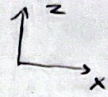


Right View



Multi View Sketch (stress) cont:

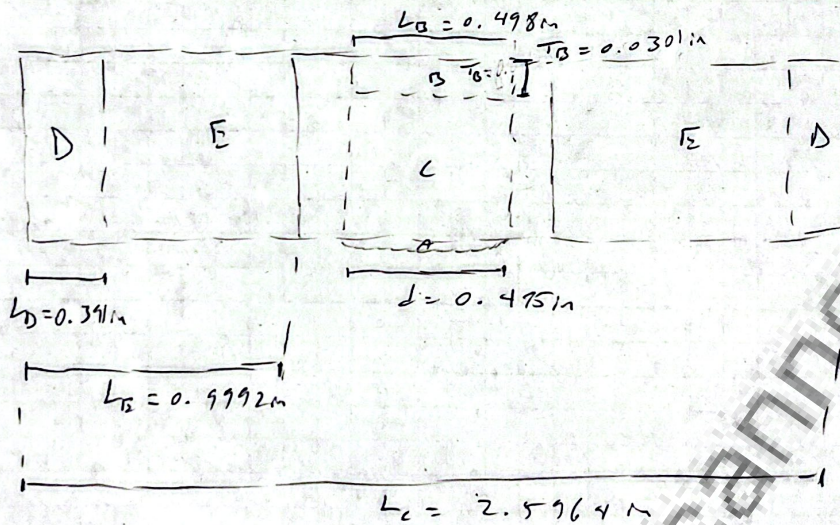
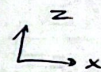
TOP View



Thickness
= 1 m
(From Assumption)

Multi View Sketch (still stress) cont:

TOP View



Thickness
= 1 m